HW #10

①

1. (a) $x(n)=[1,5,2,6]$ $h(n)=[1,0,0,1]$ $y(n)=\sum_{k=0}^{N-1} x(k)h(n-k)_N$

$n=0$, $h(-k)=[1\ 1\ 0\ 0]$ $y(0)=1\times1+5\times1=6$

$n=1$ $h(1-k)=[0\ 1\ 1\ 0]$ $y(1)=5\times1+2\times1=7$

$n=2$ $h(2-k)=[0\ 0\ 1\ 1]$ $y(2)=2\times1+6\times1=8$

$n=3$ $h(3-k)=[1\ 0\ 0\ 1]$ $y(3)=1\times1+6\times1=7$

$$\boxed{y(n)=[6,7,8,7]}$$

(b) $x(n)=[1,-1,-1,1,-1,1]$ $h(n)=[1,2,3,3,2,1]$

$n=0$, $h(-k)=[1,1,2,3,3,2]$ $y(0)=1\times1-1\times1-1\times2+1\times3-1\times3+1\times2=0$

$n=1$, $h(1-k)=[2,1,1,2,3,3]$ $y(1)=1\times2-1\times1-1\times1+1\times2-1\times3+1\times3=2$,

$n=2$ $h(2-k)=[3,2,1,1,2,3]$ $y(2)=1\times3-1\times2-1\times1+1\times1-1\times2+1\times3=2$

$n=3$ $h(3-k)=[3,3,2,1,1,2]$ $y(3)=1\times3-1\times3-1\times2+1\times1-1\times1+1\times2=0$

$n=4$ $h(4-k)=[2,3,3,2,1,1]$ $y(4)=1\times2-1\times3-1\times3+1\times2-1\times1+1\times1=-2$

$n=5$ $h(5-k)=[1,2,3,3,2,1]$ $y(5)=1\times1-1\times2-1\times3+1\times3-1\times2+1\times1=-2$

$$\boxed{y(n)=[0,2,2,0,-2,-2]}$$

2. (a) $X(k)=\sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi kn}{N}}=1+5e^{-j\frac{2\pi k}{4}}+2e^{-j\frac{2\pi k\cdot2}{4}}+6e^{-j\frac{2\pi k\cdot3}{4}}$

$=1+5e^{-j\frac{\pi k}{2}}+2e^{-j\pi k}+6e^{-j\frac{3\pi k}{2}}$

$X(0)=1+5+2+6=14$

$X(1)=1-5j-2+6j=-1+j$

$X(2)=1-5+2-6=-8$

$X(3)=1+5j-2-6j=-1-j$

$$\Rightarrow \boxed{X(k)=[14,-1+j,-8,-1-j]}$$

$$H(k) = \sum_{n=0}^{N-1} h(n) e^{-j\frac{2\pi kn}{N}}$$

$$= 1 + e^{-j\frac{2\pi k \cdot 3}{4}} = 1 + e^{-j\frac{3\pi k}{2}}$$

$H(0) = 1+1 = 2$

$H(1) = 1+j$

$H(2) = 1-1 = 0$ $\Rightarrow$ $\boxed{H(k) = [2,\ 1+j,\ 0,\ 1-j]}$

$H(3) = 1-j$

(b) From 1.(a), $y(n) = [6, 7, 8, 7]$

$$Y(k) = \sum_{n=0}^{N-1} y(n) e^{-j\frac{2\pi kn}{N}}$$

$$= 6 + 7e^{-j\frac{2\pi k}{4}} + 8e^{-j\frac{2\pi k \cdot 2}{4}} + 7e^{-j\frac{2\pi k \cdot 3}{4}}$$

$$= 6 + 7e^{-j\frac{\pi k}{2}} + 8e^{-j\pi k} + 7e^{-j\frac{3\pi k}{2}}$$

$Y(0) = 6+7+8+7 = 28$

$Y(1) = 6-7j-8+7j = -2$

$Y(3) = 6-7+8-7 = 0$

$Y(4) = 6+7j-8-7j = -2$ $\Rightarrow$ $\boxed{Y(k) = [28, -2, 0, -2]}$

$X(k) \cdot H(k) = [14.2,\ (1+j)(1+j),\ 0.(-8),\ (1-j)(-1-j)]$

$\boxed{= [28,\ -2,\ 0,\ -2]}$

So that $Y(k) = X(k) \cdot H(k)$

3. Use the property: $x(n)\cdot h(n) \Longleftrightarrow \frac{1}{N}\sum_{k=0}^{N-1} X(k)\,H(n-k)_N$ (3)

$x(n)\cdot h(n) = [1,5,2,3,6]\cdot[2,3,0,3,2] = [2,15,0,9,12]$

$y(n) = 5\cdot x(n)\cdot h(n) = [10,75,0,45,60]$

$$Y(k) = \sum_{n=0}^{N-1} y(n) e^{-j\frac{2\pi kn}{N}}$$

$$= 10 + 75e^{-j\frac{2\pi k}{5}} + 45e^{-j\frac{2\pi k\cdot 3}{5}} + 60e^{-j\frac{2\pi k\cdot 4}{5}}$$

$$= 10 + 75e^{-j\frac{2\pi k}{5}} + 45e^{-j\frac{6\pi k}{5}} + 60e^{-j\frac{8\pi k}{5}}$$

Using matlab: $Y(k) = [190,\ 15.31+12.18j,\ -85.31-51.61j,\ -85.31+51.61j,\ 15.31-12.18j]$

4. Use the time shifting property:

$$x(n-n_0)_N \longleftrightarrow X(k)\,e^{-j\frac{2\pi kn_0}{N}}$$

$y(n) = x(n-2)_6 \longleftrightarrow Y(0) = 5$

$$Y(1) = 2e^{-j\frac{2\pi\cdot 1\cdot 2}{6}} = 2e^{-j\frac{2\pi}{3}} = -1-\sqrt{3}j$$

$$Y(2) = 3\cdot e^{-j\frac{2\pi\cdot 2\cdot 2}{6}} = 3e^{-j\frac{4\pi}{3}} = -\frac{3}{2}+\frac{3\sqrt{3}}{2}j$$

$$Y(3) = 1e^{-j\frac{2\pi\cdot 3\cdot 2}{6}} = e^{-j2\pi} = 1$$

$$Y(4) = 0$$

$$Y(5) = 4e^{-j\frac{2\pi\cdot 5\cdot 2}{6}} = 4e^{-j\frac{10\pi}{3}} = -2+2\sqrt{3}j$$

$$Y(k) = [5,\ -1-\sqrt{3}j,\ -\tfrac{3}{2}+\tfrac{3\sqrt{3}}{2}j,\ 1,\ 0,\ -2+2\sqrt{3}j]$$